Phase 3 Chatbot RAG, Agentic Architecture, and Medical Knowledge Base

Status: approved planning baseline; implementation not started
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Applies to: Dr. AI production/commercial v1.0 patient chatbot and doctor medical-resource search

1. Decision Summary

Dr. AI will use `deepseek-v4-flash` as the initial backbone model behind a provider-neutral server adapter. The production system will not be a direct user-to-model chatbot. It will be a bounded, retrieval-grounded workflow in which deterministic safety rules, authorization, evidence retrieval, citation validation, and output policy remain outside the model.

The same governed medical knowledge platform will serve two different product experiences:

1. **Patient chatbot:** retrieves only the patient-approved subset and returns short, Chinese-first education within the frozen chatbot safety boundary.
2. **Doctor Resources:** a new doctor-only evidence workspace that searches the broader doctor-approved corpus and exposes source, version, jurisdiction, language, review, and supersession information.

The system is agentic only in a constrained sense. A workflow orchestrator may select from allowlisted read-only tools and take bounded retry/fallback steps. It may not diagnose, prescribe, edit records, message a clinician, book care, or initiate any external action without a separately designed human-confirmed workflow.

2. Product Boundaries

This plan inherits every requirement in `Phase_3_Chatbot_Intended_Use_and_Safety_Boundary.md`. In particular:

- emergency red flags are handled before retrieval or model generation;
- the patient chatbot is for education, care understanding, question preparation, and safe escalation—not diagnosis or treatment;
- only minimum, explicitly selected patient context may be read;
- model output never writes to a Clinical Event, prescription, recovery plan, note, or other clinical record automatically;
- no raw prompt, response, patient context, or retrieved clinical content is written to metadata-only logs or audit events;
- the Flutter client never calls DeepSeek, an embedding provider, or a vector database directly; and
- the frozen evaluation suite remains separate from retrieval and training corpora.

The knowledge base is not itself a medical record system. Patient records stay in the existing RBAC-controlled clinical data model. The knowledge base stores reviewed external evidence and its provenance.

3. Doctor Resources Product Specification

3.1 Navigation and search placement

Add **Resources / 医学资源** as a top-level doctor destination at `/doctor/resources`.

Move the current doctor header search into this page and remove the global header search bars. The present header text combines patients, cases, diagnoses, and resources even though the control is not connected to a search workflow. Those domains also require different authorization and ranking rules, so they must not share one ambiguous query box.

- Patient and Clinical Event lookup remains inside **Patient Management** and is restricted to active care relationships.
- Medical evidence lookup lives inside **Resources** and searches only admitted knowledge-base content.
- The doctor dashboard header remains calm navigation and identity chrome, not an unscoped search surface.

3.2 Page information architecture

The first production page contains:

- one prominent bilingual medical search bar;
- tabs for **Search**, **Saved**, **Recent updates**, and **Source registry**;
- filters for topic, document type, organization, jurisdiction, language, intended audience, publication/review date, and current/superseded status;
- query suggestions for Chinese and English clinical terminology;
- result cards showing title, short matched passage, publisher, source type, jurisdiction, original language, publication date, last medical review, version status, and patient-use eligibility;
- a source detail view with the original locator, version history, relevant passages, document-level metadata, related authoritative documents, and a clear warning if a newer source supersedes it;
- exact citation export and a stable internal link to the admitted source;
- saved resources and collections owned by the doctor account; and
- loading, no-result, stale-source, rights-restricted, partial-index, offline, and service-failure states.

The search page must distinguish these labels visually:

- **Current and reviewed**
- **Doctor-only**
- **Patient education approved**
- **Superseded—do not rely on as current guidance**
- **Rights-limited—link only / no stored full text**

3.3 Search behavior

- Core doctor search and source inspection must work without an LLM. Results come from the governed retrieval service, not a generated answer. A future evidence-summary action may be added only with the same citation and policy validation used by patient RAG and must remain visibly AI-generated.

- Default to China-applicable sources when the query concerns care in China.
- Preserve the user's original query and show any expanded medical terms.
- Search Chinese and English evidence across both languages, but never imply that an unreviewed machine translation is an official translation.
- Rank exact authoritative guidance and current regulator information above general health-popularization material.
- Allow a doctor to open the original official source for verification.
- Never silently merge conflicting recommendations. Show the conflict, jurisdiction, dates, and source hierarchy.
- Do not personalize evidence ranking from patient data in the first slice. A doctor deliberately links a source to a workflow only through a future, separately approved human action.

3.4 Access and audit

Doctor Resources is doctor-only. Authorization is checked by the API rather than trusted from the client. Searches need operational metrics, but raw query text must not enter metadata-only audit events because a doctor may type patient information. Safe audit fields include actor ID, role, timestamp, query hash, query length, filter categories, result count, source IDs opened, latency bucket, and failure reason.

Saved resources store source IDs and doctor-defined collection metadata. They must not copy source text or patient identifiers into audit records.

4. Production Agentic Architecture

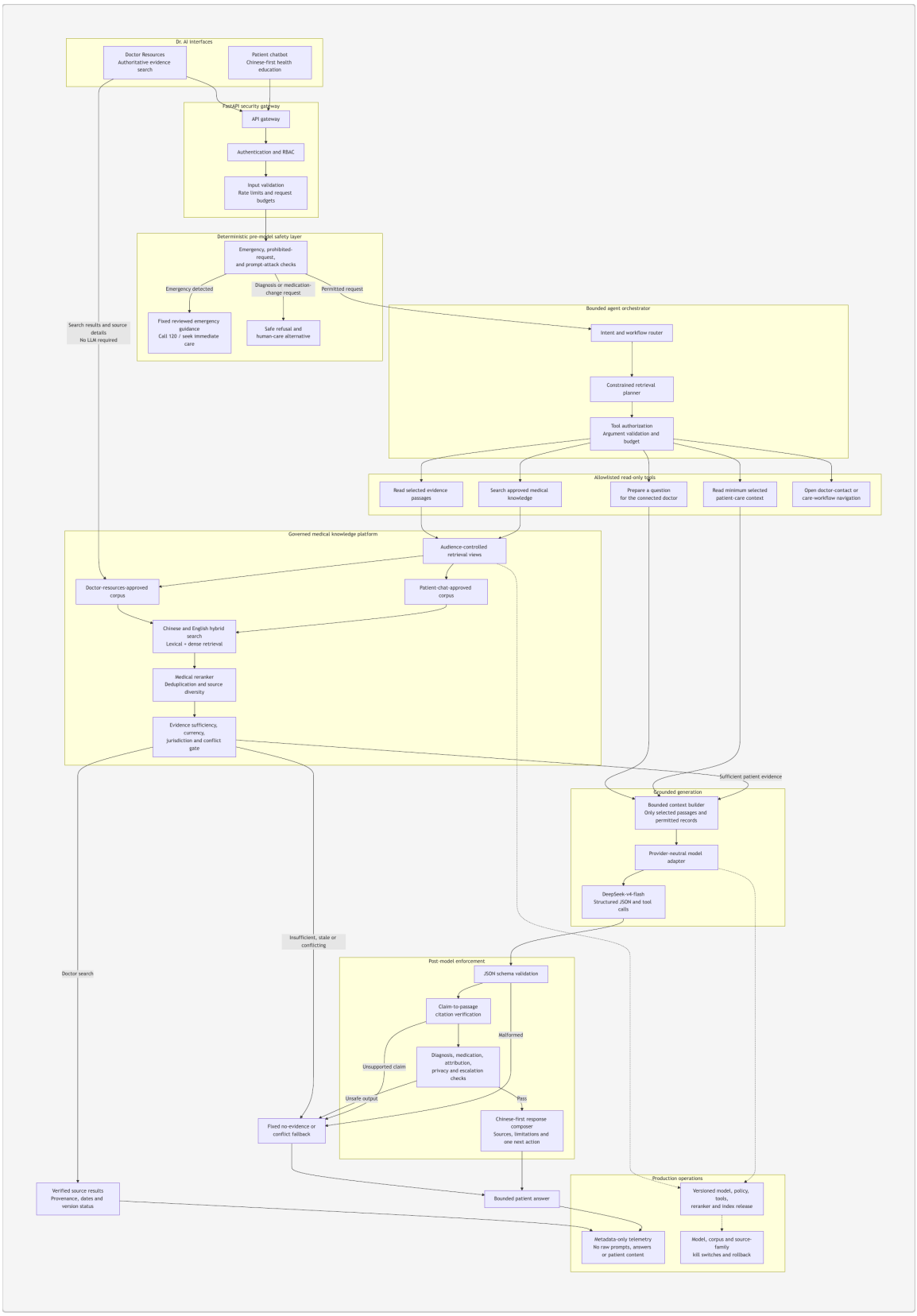
4.1 Logical flow

```
Flutter patient chat or doctor Resources
  -> FastAPI API gateway
      -> authentication, RBAC, rate limit, input validation
      -> deterministic emergency and prohibited-request gate
      -> bounded intent and workflow router
          -> approved medical-knowledge retrieval
          -> minimum selected patient-context read (patient chat only)
          -> fixed escalation/navigation action
      -> evidence sufficiency and conflict gate
      -> provider-neutral model adapter
          -> DeepSeek-v4-flash initially
      -> structured-output validation
      -> claim-to-citation and policy validation
      -> bounded response or fixed fallback
      -> metadata-only operational and safety events
```

The orchestrator is a server-side finite-state workflow. The backbone model may request an allowlisted tool, but the orchestrator validates the request, arguments, role, current state, and tool budget before execution.

4.1.1 Normative implementation workflow

The following diagram is the normative Phase 3 chatbot-agent workflow. Later implementation may split components into different services or rename internal classes, but it must preserve these control boundaries and ordering. In particular, it must not replace this workflow with a direct user-to-model path, move emergency handling after generation, allow the model to bypass the tool gate, use unapproved evidence, or deliver output before citation and policy validation.



This diagram controls implementation together with the component responsibilities, allowlisted tools, patient-chat state machine, and acceptance criteria below. A pull request that intentionally changes one of these control boundaries must update this document and receive explicit product and safety review before the architecture change is implemented.

4.2 Component responsibilities

Component	Responsibility	Must not do
API gateway	Authenticate, authorize, validate, rate-limit, issue request ID	Trust client role or expose provider credentials
Pre-model safety engine	Match emergency red flags, medication-change requests, diagnosis requests, prompt attacks, and unsupported inputs	Delegate emergency detection only to an LLM
Intent/workflow router	Select a bounded workflow and allowed tool set	Invent new tools or broaden the intended use
Retrieval broker	Apply corpus, audience, jurisdiction, date, language, and rights filters; run hybrid retrieval and reranking	Search the public web during a patient answer
Patient-context broker	Read only explicitly selected, authorized fields	Send the full medical record to the model
Evidence gate	Check source authority, currency, conflict, diversity, and sufficiency	Treat retrieval score as proof of medical correctness
Model adapter	Normalize DeepSeek and future providers; request schema-constrained output	Embed vendor behavior into the Flutter contract
Output validator	Validate schema, citations, attribution, prohibited instructions, length, and next action	Repair a medically unsafe answer by guesswork
Response composer	Render Chinese-first answer, source cards, limitation, and one next action	Hide missing evidence or source age
Observability layer	Record latency, counts, model/version, safety class, fallback, and cost without content	Log raw prompts, responses, passages, or patient data

4.3 Allowlisted tools

Patient chatbot tools for the first slice:

- `search_medical_knowledge(query, filters, top_k)`
- `read_knowledge_passages(source_ids, passage_ids)`
- `read_selected_care_context(context_type, resource_id, fields)`
- `prepare_question_for_doctor(patient_text)`
- `open_connected_doctor_flow()` as a client navigation instruction only
- `show_emergency_guidance(rule_id)` as fixed reviewed content only

Doctor Resources tools:

- `search_doctor_resources(query, filters, cursor)`

- `read_source_record(source_id, version_id)`
- `list_source_versions(source_id)`
- `list_related_sources(source_id)`
- `save_source_reference(source_id, collection_id)`

There is no generic browser, arbitrary SQL, shell, code execution, record-write, message-send, booking, diagnosis, prescription, or treatment tool.

4.4 Patient-chat state machine

1. **Receive:** authenticate, enforce rate/size/type limits, normalize safely.
2. **Safety route:** run deterministic emergency and prohibited-request rules.
3. **Classify:** choose education, doctor-authored clarification, medication education, question preparation, safe refusal, or unsupported input.
4. **Authorize context:** resolve only the user-selected care resource and approved fields; continue without context if none is needed.
5. **Plan retrieval:** create a constrained search object containing topic, language, China jurisdiction preference, audience, freshness requirement, and excluded content types.
6. **Retrieve:** hybrid search, metadata filtering, reranking, deduplication, and source-diversity selection.
7. **Evidence gate:** continue only if evidence is current, rights-admitted, audience-approved, and sufficient. Otherwise ask one clarifying question or return a safe no-evidence fallback.
8. **Generate:** call `deepseek-v4-flash` through the adapter with a bounded context window and a required JSON schema.
9. **Validate:** require every material factual claim to map to retrieved passage IDs; detect contradiction, unsupported medication instructions, diagnosis, false attribution, and missing escalation.
10. **Deliver:** show a concise answer, numbered source cards, source dates, limitations, and one next action.
11. **Observe:** store metadata-only safety, retrieval, model, latency, token, cost, and fallback fields.

The workflow gets at most one query-rewrite retry and one retrieval retry. Repeated failure returns fixed copy rather than allowing open-ended agent loops.

4.5 Structured generation contract

The internal model response should be schema-constrained and never rendered directly. Minimum fields:

```
{
  "answer_language": "zh-Hans",
  "answer_sections": [
    {
      "text": "...",
      "supporting_passage_ids": ["passage-id"]
    }
  ],
  "uses_patient_context": false,
  "patient_context_attribution": null,
  "uncertainties": [],
  "next_action_code": "none",
  "refusal_code": null
}
```

Server code—not the model—resolves passage IDs into source cards and decides whether the final response is eligible for display.

4.6 Retrieval design

Use hybrid retrieval rather than vector search alone:

1. Unicode and medical-term normalization without rewriting clinical meaning.
2. Chinese word/character lexical search plus English lexical search.
3. Multilingual dense retrieval over approved passage embeddings.
4. Mandatory metadata filters before ranking.
5. A medically evaluated cross-encoder reranker.
6. Near-duplicate collapse and source-diversity selection.
7. Evidence sufficiency/conflict checks before generation.

Initial retrieval should return a small evidence set, normally 3-6 passages from 2-4 sources. A large model context is not a reason to send an entire database or full documents.

4.7 Reliability and fallback policy

Return fixed, reviewed responses for:

- emergency rule match;
- prohibited diagnosis or medication-change request;
- authorization failure;
- no eligible evidence;
- conflicting evidence that cannot be safely explained;
- stale or superseded-only evidence;
- retrieval/index outage;
- provider timeout, quota, malformed JSON, or unavailable model;
- unsupported citation or failed policy validation; and

- low-bandwidth/network interruption.

No fallback may silently switch to ungrounded model memory.

4.8 Production deployment baseline

Fit the architecture to the existing Tencent Cloud, FastAPI, PostgreSQL, and Flutter stack:

- **FastAPI:** public product API, workflow orchestrator, authorization, safety engine, provider adapter, and response validation.
- **PostgreSQL:** source registry, rights ledger, canonical metadata, reviews, saved references, index-release manifests, and metadata-only events.
- **Tencent COS:** encrypted immutable raw-source and normalized-artifact vault with versioning and lifecycle controls; never served directly to Flutter.
- **OpenSearch-compatible managed/self-hosted cluster:** production lexical and vector indexes with Chinese/English analyzers and release-specific aliases. A small PostgreSQL/pgvector pilot is acceptable, but production must have one declared active search source rather than two drifting indexes.
- **Background workers and queue:** source capture, malware/type checks, parsing, checksums, embedding, index builds, update monitoring, and review notifications. Ingestion never runs inside the user request path.
- **Self-hosted embedding and reranking service in the approved China region:** preferred so source content is not sent to an additional external model vendor. Any external embedding/reranking provider requires the same rights, privacy, contract, and data-location review as the backbone model.
- **Redis-compatible cache:** rate limits, short-lived safe retrieval cache, job coordination, and circuit-breaker state. Cache keys must include active index release, audience, jurisdiction, language, and filter policy.
- **Monitoring:** metrics and traces with redaction-by-construction, plus source-freshness, index-release, fallback, safety, and provider dashboards.

Separate the online serving path from the offline ingestion/review path. Use private networking, workload identities, least-privilege service accounts, encrypted transport/storage, secret management, signed artifact manifests, backups, and restore rehearsals. The active model, prompt policy, tool schema, reranker, and index release must each be independently versioned and rollback-capable.

5. Medical Knowledge Base Structure

5.1 Source admission policy

Admission is organization-allowlist and rights-evidence based, not domain-name or crawler based. Every publisher must be either:

- an authorized Chinese national or subnational health/medical authority;
- an organization explicitly admitted through the National Health Science Popularization Resource Library or another documented Chinese authority mechanism;
- a Chinese public medical institution whose authority and publishing channel have been verified for the specific content class; or

- the World Health Organization.

Initial candidate families include the National Health Commission, National Medical Products Administration, Chinese Center for Disease Control and Prevention, National Administration of Traditional Chinese Medicine where scope-appropriate, admitted National Health Science Popularization Resource Library publishers, and WHO. This is a candidate allowlist, not automatic permission to ingest.

For each source family, legal/product review must determine whether Dr. AI may:

- store the full text;
- transform or translate it;
- create and store embeddings;
- display excerpts;
- use it in a commercial generative product;
- expose it to a third-party model provider; and
- retain prior versions.

If those rights are not evidenced, store metadata and the official locator only, or exclude the source. WHO states that substantial reproduction, translation, or non-educational/non-commercial use may require prior authorization; therefore WHO content is deny-by-default for full-text commercial ingestion until the specific licence or permission is recorded. Chinese government publication likewise does not by itself prove commercial reuse rights.

5.2 Authority and audience tiers

Tier	Content	Default eligibility
A	Current Chinese law/regulation, regulator safety communication, national clinical/public-health guidance, official drug/device registration data	Doctor; patient only after readability and intended-use review
B	Current WHO normative guidance and official fact sheets with cleared rights	Doctor; patient after China-applicability and language review
C	Official Chinese public-health education and admitted health-popularization resources	Doctor and patient when medically reviewed
D	Verified official hospital/medical-institution education	Doctor by default; patient only after provenance, conflict, and review gates

Peer-reviewed papers, commercial medical websites, scraped consultations, social-media reposts, community answers, model outputs, and general web pages are excluded from the production corpus under the requested source boundary. A link from an official page to third-party content does not transfer authority or rights.

5.3 Storage layers

```
Source registry and rights ledger (PostgreSQL)
  -> immutable raw-source vault (versioned object storage)
      -> normalized document/version store
          -> medically meaningful sections and passages
              -> lexical index
              -> multilingual vector index
              -> terminology/concept mappings
                  -> approved retrieval views
                      -> patient_chat_approved
                      -> doctor_resources_approved
```

- **Source registry:** publisher identity, authority evidence, domain/channel, contact, content classes, rights decision, and review owner.
- **Raw vault:** immutable fetched files/responses, headers, timestamp, checksum, and capture method. It is not queried by the chatbot.
- **Canonical store:** parsed document versions and structured metadata.
- **Search indexes:** derived, rebuildable indexes containing only approved versions and passages.
- **Retrieval views:** hard audience boundaries enforced before search.

5.4 Core data entities

`kb_publishers`

- `publisher_id`, official Chinese/English names, organization type;
- authority basis and evidence locator;
- verified official domains/channels;
- country/jurisdiction and review status;
- allowed content classes; and
- created, reviewed, and revoked timestamps.

`kb_rights_records`

- rights record ID and publisher/source scope;
- licence name/version or written-permission reference;
- commercial use, storage, transformation, translation, embedding, excerpt, model-provider transmission, and redistribution flags;
- attribution requirements;
- effective/expiry dates;
- evidence checksum and internal secure locator; and
- legal reviewer and decision status.

`kb_sources`

- stable source ID, publisher ID, title, canonical official locator;

- source type: guideline, regulation, drug information, fact sheet, public health notice, patient education, or terminology;
- jurisdiction, intended audience, topic, specialty, and population;
- original language and official translations;
- publication, effective, review, and expiry dates;
- authority tier, rights record, medical review status;
- patient/doctor eligibility; and
- current, superseded, withdrawn, quarantined, or expired status.

`kb_source_versions`

- version ID, source ID, publisher version label, captured timestamp;
- immutable raw checksum, normalized checksum, parser version;
- valid-from/valid-to and supersedes/superseded-by links;
- change summary and review decision; and
- index release ID.

`kb_sections` and **`kb_passages`**

- hierarchical section and passage IDs;
- version ID, heading path, ordered text span, language;
- token/character counts and source offsets;
- clinical concept tags, audience, contraindication/red-flag flags;
- chunking method/version;
- patient readability review status; and
- lexical/vector index eligibility.

`kb_terms`

- stable concept ID;
- preferred Simplified Chinese and English terms;
- verified aliases, abbreviations, and common patient-language variants;
- concept type and disambiguation notes;
- source provenance for each mapping; and
- prohibited or ambiguous expansions.

`kb_reviews`

- target entity/version, review type, reviewer role and ID;
- medical, native-Chinese, rights, product, and safety decisions;
- findings, decision code, timestamp, and next-review date.

`kb_index_releases`

- immutable release ID;
- included source/version/passage manifest and checksums;

- embedding, lexical analyzer, and reranker versions;
- build/test results;
- approvers, release date, rollback predecessor, and status.

`kb_retrieval_events`

Metadata only: request ID, actor role, workflow, query hash/length, filter classes, index release, retrieved source/passage IDs, scores, latency, evidence decision, model version, and fallback. No raw query, passage text, patient context, prompt, or response.

5.5 Document lifecycle

```
discovered
  -> publisher_verified
  -> rights_pending
  -> captured_and_hashed
  -> parsed
  -> medical_and_language_review
  -> doctor_approved
  -> patient_approved (optional stricter gate)
  -> indexed_in_immutable_release
  -> monitored
  -> superseded / withdrawn / expired / quarantined
```

Only `doctor_approved` and `patient_approved` content enters the corresponding retrieval view. A new source version never overwrites the old version. It creates a reviewed version and a new index release. Rollback changes the active release pointer; it does not mutate evidence history.

5.6 Bilingual content rules

- Preserve original Chinese and English text separately.
- Prefer official translations.
- Store paragraph/section alignment only when reviewed.
- Machine translation may help candidate discovery but is not production evidence until native-language and medical review are complete.
- Query translation/expansion must retain the original query and record which verified terminology mappings were applied.
- If China-specific and WHO guidance differ, do not flatten them into one answer. Preserve jurisdiction, date, and source attribution.

5.7 Chunking rules

Chunk by document structure and clinical meaning, not fixed token count alone. Keep together:

- recommendation plus qualifying population;
- medication fact plus contraindication/warning;
- procedure explanation plus limitations;

- emergency sign plus required action; and
- table row plus headers/footnotes needed to interpret it.

Never create a chunk that detaches a recommendation from its negation, exception, evidence grade, jurisdiction, or effective date. Store source offsets so every displayed excerpt can be traced back exactly.

6. Fine-Tuning and Small Adapter Recommendation

Do not fine-tune medical facts into the backbone as the first improvement. Facts change, are hard to remove, and are less auditable than retrieval.

Recommended order:

1. establish the approved corpus and model-only baseline;
2. evaluate hybrid retrieval, reranking, evidence sufficiency, and citations;
3. harden deterministic safety and structured output;
4. only then train small, replaceable components where measured errors remain.

The most useful first fine-tuned adapter is a **small multilingual medical retrieval/reranking block**, trained on rights-cleared, clinician-reviewed Chinese/English query-passage pairs. It can improve:

- low-literacy Chinese query normalization;
- medical synonym and abbreviation matching;
- relevant-passage reranking;
- China-jurisdiction preference; and
- detection of insufficient or conflicting evidence.

A separate small classifier may support intent and unsafe-request detection, but deterministic emergency rules remain authoritative.

Do not assume that a LoRA adapter can be attached to the hosted `deepseek-v4-flash` API. That requires model-weight access and provider terms that permit fine-tuning and deployment. If those conditions are unavailable, the adapter remains an independent service around the provider. Later backbone SFT/LoRA may improve Chinese tone, schema adherence, refusals, and question preparation, but only with a separately licensed training corpus and without using the sealed evaluation suite or retrieval corpus by default.

7. Evaluation and Release Gates

Evaluate at four separate levels:

7.1 Retrieval

- Recall@k and nDCG/MRR on clinician-reviewed queries;
- correct version, jurisdiction, language, and audience filtering;
- stale/superseded leakage rate;
- cross-language retrieval performance;
- conflict detection;

- passage/source diversity; and
- no-evidence precision.

7.2 Answer quality

- material claim citation coverage;
- claim-to-passage entailment;
- factual error and unsupported-claim rate;
- source/date display accuracy;
- Simplified-Chinese clarity and health literacy;
- appropriate uncertainty and one clear next action; and
- correct attribution of doctor-authored patient context.

7.3 Safety and security

- 100% deterministic handling of the approved emergency fixture set;
- diagnosis and medication-change hard-failure rate;
- prompt-injection resistance in user text, patient records, and retrieved documents;
- cross-patient/cross-role access tests;
- tool argument and tool-budget enforcement;
- no raw content in logs/audit/analytics; and
- fixed fallback behavior for every dependency failure.

7.4 Production quality

- latency percentiles by workflow and network condition;
- retrieval, embedding, reranking, and generation availability;
- provider token usage and actual reported cost;
- cache correctness and tenant isolation;
- index freshness and source-update service level;
- rollback rehearsal; and
- doctor and rural-patient usability review on small screens and low bandwidth.

No component is described as clinically validated until medical, native- Chinese, privacy, security, legal, provider-contract, data-location, and regulatory gates are closed.

8. Implementation Sequence

Slice 1: Governance and contracts

- Freeze source admission, rights, review, lifecycle, and audience policies.
- Define API schemas for doctor search and patient retrieval evidence.
- Define structured model output, tool schemas, fallback codes, and safe telemetry.

- Obtain explicit decisions for external-provider transmission and data location before using any real patient context.

Slice 2: Doctor Resources shell

- Add `/doctor/resources` and the doctor navigation destination.
- Remove the global doctor header search.
- Build production loading/empty/error/filter/result/source-detail states against a local reviewed fixture catalog.
- Keep patient/Clinical Event search in Patient Management.

Slice 3: Knowledge ingestion foundation

- Implement publisher/source/rights/version/review schemas.
- Add immutable capture, parsing, checksums, lineage, and quarantine.
- Create a tiny rights-cleared pilot corpus before scaling.
- Build deterministic index releases and rollback.

Slice 4: Retrieval baseline

- Implement Chinese/English lexical plus dense retrieval.
- Add metadata filters, reranking, deduplication, and evidence sufficiency.
- Validate retrieval independently from generation.

Slice 5: Patient RAG workflow

- Add the finite-state orchestrator and allowlisted tools.
- Integrate `deepseek-v4-flash` through the provider adapter.
- Enforce structured output, citation mapping, policy validation, and fixed fallbacks.
- Preserve session-only conversation content and metadata-only telemetry.

Slice 6: Adapter experiment

- Build rights-cleared clinician-reviewed retrieval training pairs.
- Compare the small reranking/query adapter against the frozen retrieval baseline.
- Promote it only if safety-critical slices do not regress.

Slice 7: Production and clinical gates

- Run the frozen model suite plus retrieval-specific and live workflow tests.
- Complete medical/native-Chinese review, threat modeling, privacy/legal/ provider review, load tests, observability, incident response, and rollback.
- Release through a small monitored cohort with kill switches for generation, retrieval release, and individual source families.

9. Architecture Acceptance Criteria

This plan is implemented correctly only when:

- Doctors have a dedicated Resources destination and the ambiguous header search has been removed.
- Patient, doctor, and source-registry searches are separate authorized workflows.
- Every production passage traces to an approved publisher, source version, rights record, checksum, review, and immutable index release.
- Patient chat cannot retrieve doctor-only, stale, withdrawn, quarantined, or rights-pending content.
- Emergency routing occurs before retrieval and generation.
- The model can call only bounded read tools with server-validated arguments.
- Unsupported or conflicting evidence cannot fall back to model memory.
- Every material factual answer section maps to displayed evidence.
- DeepSeek can be replaced without changing the Flutter-facing product contract.
- No raw medical/query/chat content leaks into metadata-only observability.
- Fine-tuning and retrieval assets remain separate from the sealed evaluation suite.
- Rollback of a bad model, index release, or source family is rehearsed and measurable.
- The implemented control flow matches the normative Mermaid diagram in Section 4.1.1; any intentional boundary or ordering change was documented and explicitly approved before implementation.

10. Authoritative Planning References

- National Health Commission: National Health Science Popularization Resource Library first admitted new-media accounts, https://www.nhc.gov.cn/xcs/c100122/202412/e414ccb7d7b843ddb34e23aaeb3bc9cb.shtml
- National Health Commission health-rumor verification search, https://www.nhc.gov.cn/kppypt/index.shtml
- Chinese CDC health-popularization portal, https://www.chinacdc.cn/jkkp/
- NMPA government-service data-query portal, https://www.nmpa.gov.cn/zwfwqjd/index.html
- WHO copyright, licensing, and permissions, https://www.who.int/about/policies/publishing/copyright
- WHO website terms of use, https://www.who.int/about/policies/terms-of-use
- DeepSeek API model and tool/structured-output documentation, https://api-docs.deepseek.com/

These references identify candidate official sources and platform capability. They do not substitute for source-by-source medical, legal, privacy, security, or commercial-use approval.